Open-Source Analog IC Design Workshop

1.3 Open-source analog design tools

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ESCUELA DE INGENIERÍA ELÉCTRICA — PONTIFICIA UNIVERSIDAD CATÓLICA DE VALPARAÍSO

AC3E

SEED: SEMICONDUCTOR EDUCATION, EMPOWERMENT, & INDUSTRY DEVELOPMENT

IEEE Solid-State Circuits Society IC Innovation — Electron Devices Society

Agenda

- Introduction
- Open-source analog design tools
- Simulation examples

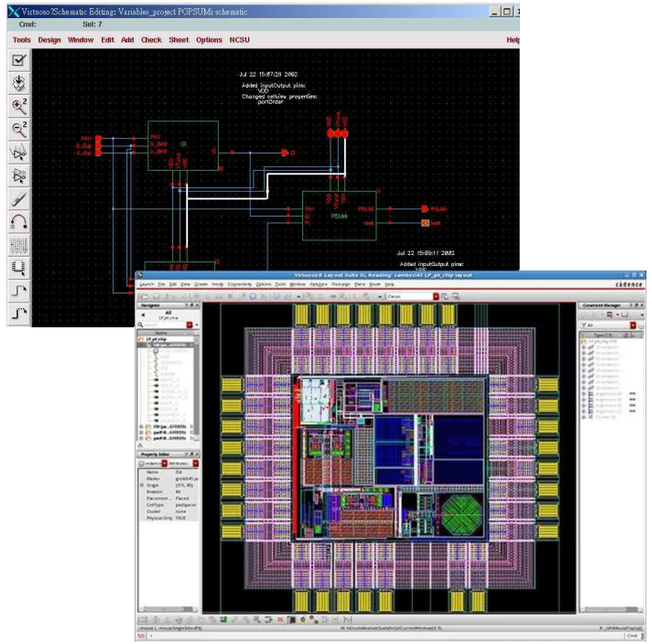
Introduction

Introduction

IC design: from manual to automated



< 1990s → fully manual chip design!



> 1990s → EDA tools

Google + skywater TECHNOLOGY

FOSS 130nm Production PDK
github.com/google/skywater-pdk

IHP-GmbH/**IHP-Open-PDK-docs**

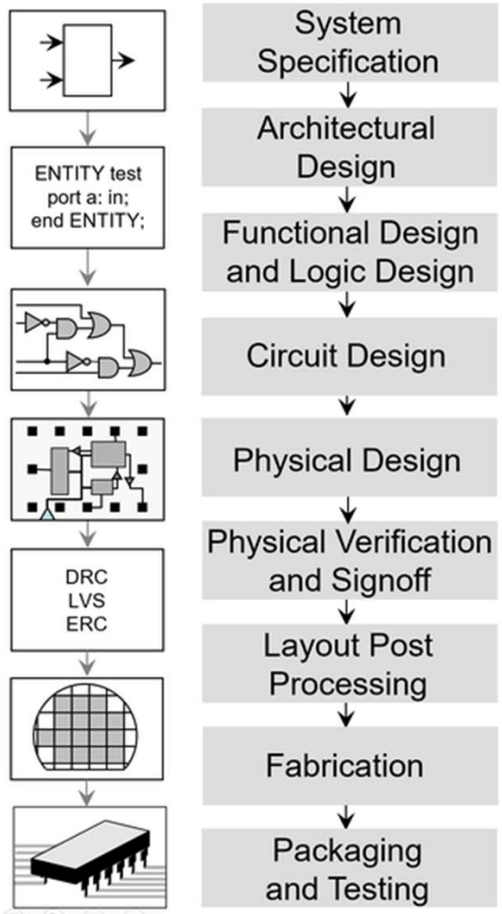
Documentation for IHP 130nm BiCMOS Open Source PDK

> 2020 → Opensource EDA tools and PDKs

Analog vs digital design flows

IC design flow: digital versus analog

DIGITAL FLOW

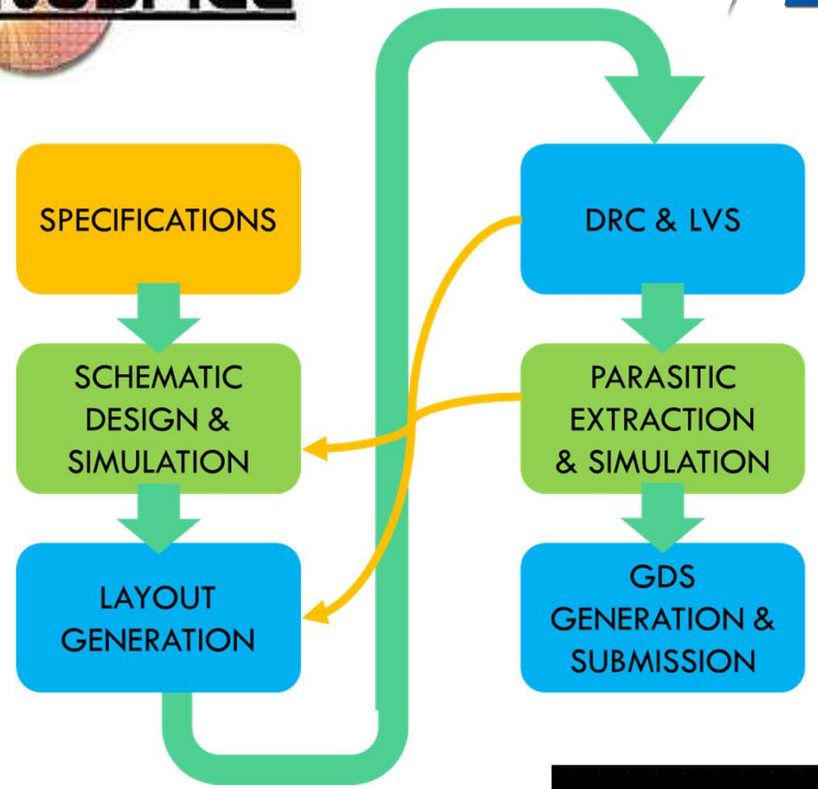


System Specification
↓
Architectural Design
↓
Functional Design and Logic Design
↓
Circuit Design
↓
Physical Design
↓
Physical Verification and Signoff
↓
Layout Post Processing
↓
Fabrication
↓
Packaging and Testing

ANALOG FLOW

NGSPICE

Xyce PARALLEL ELECTRONIC SIMULATOR



XSCHEM

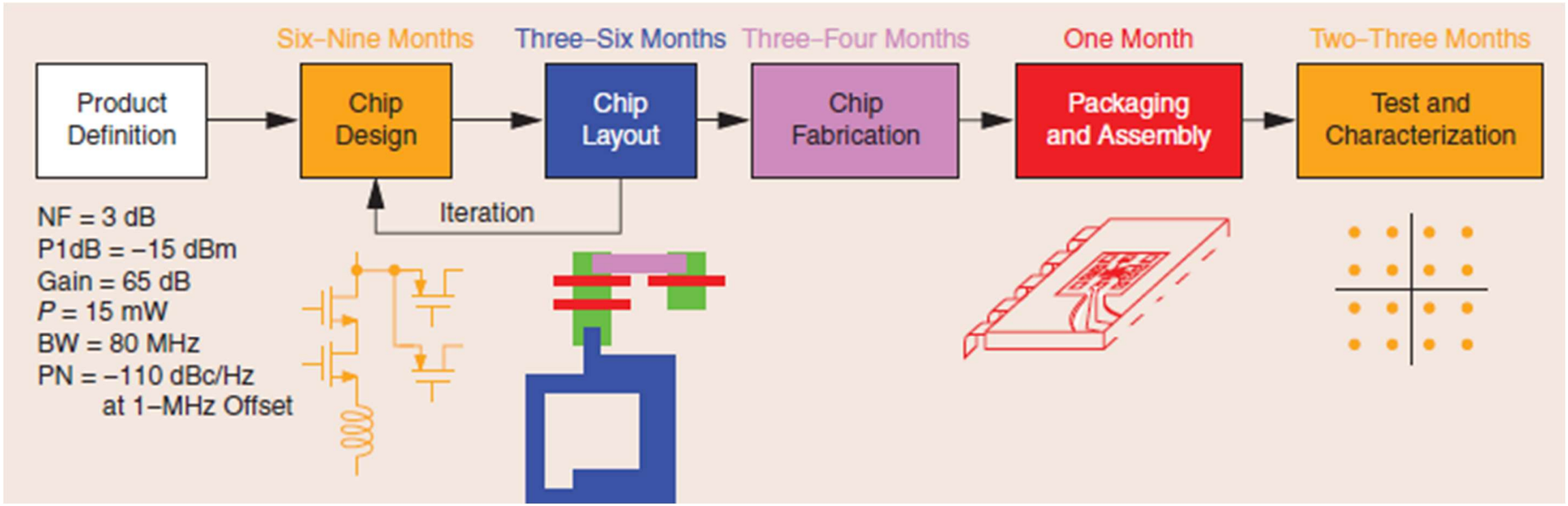
Yosys

OpenROAD 24h

Magic VLSI Layout Tool
Current distribution version 8.2

Introduction

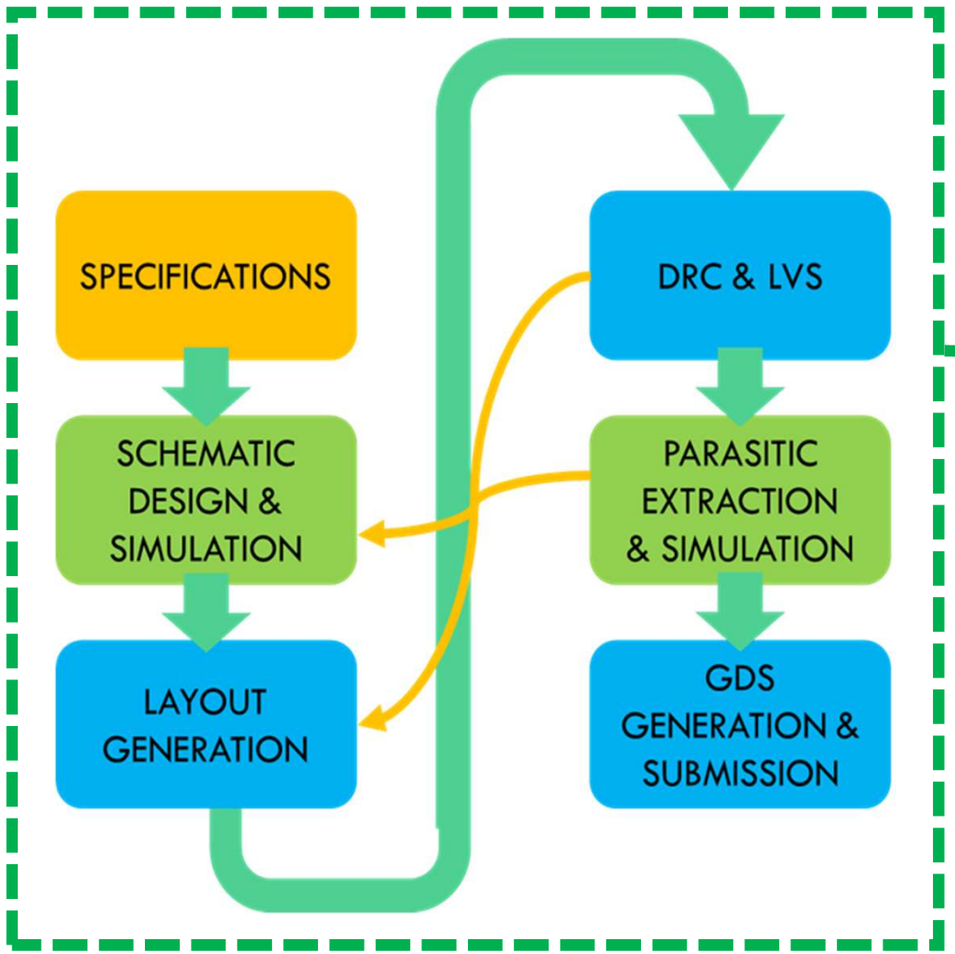
Silicon-proven analog IC design

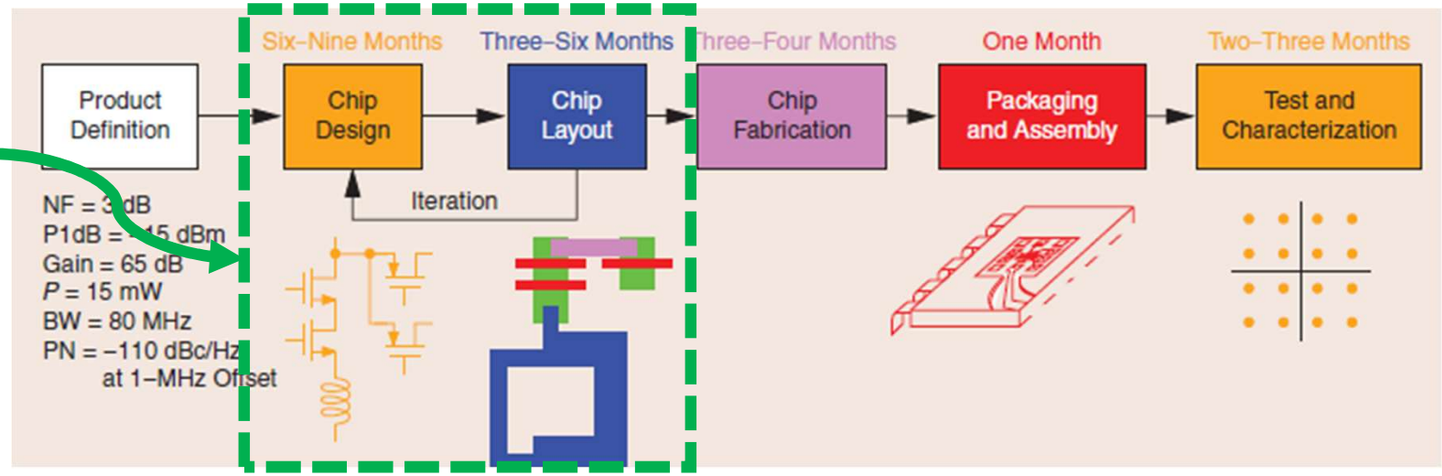


[RAZAVI, IEEE SSCMAG 2024]

Introduction

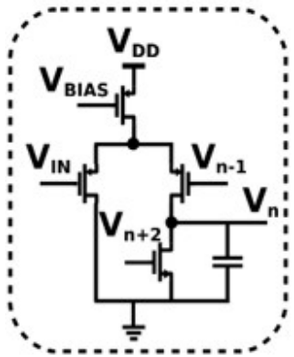
Silicon-proven analog IC design

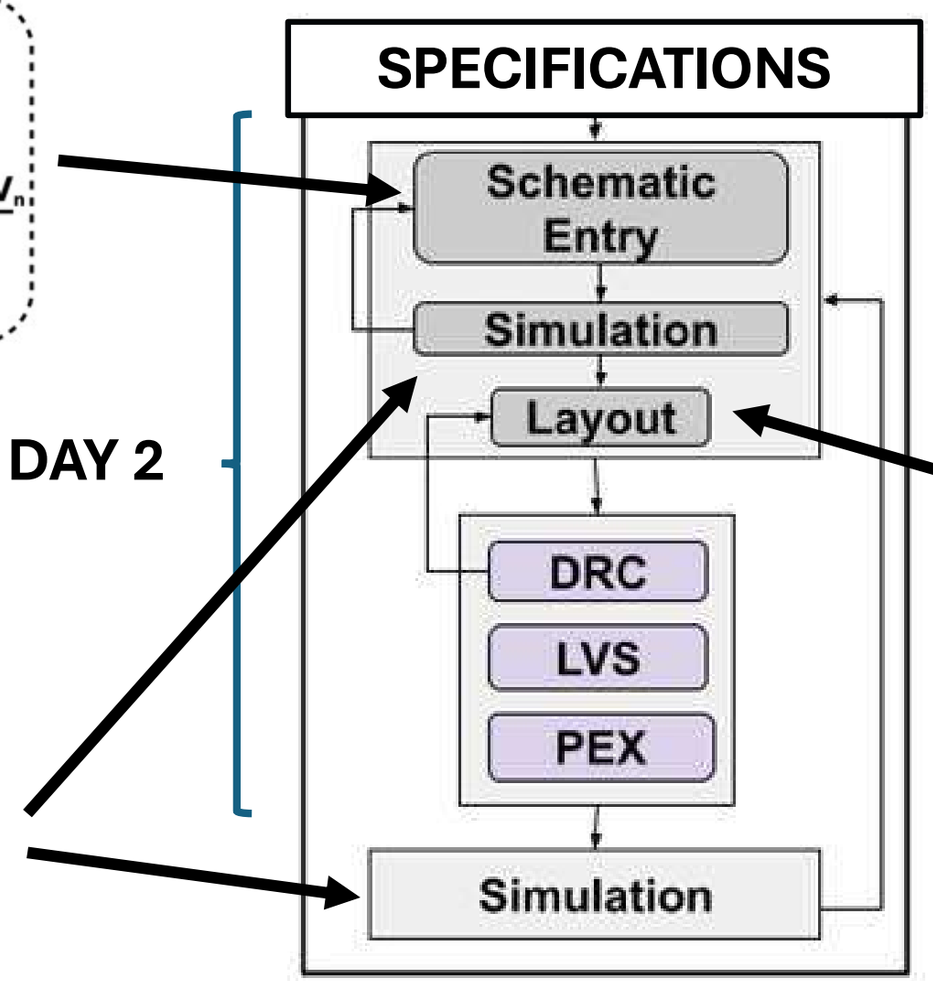




Introduction

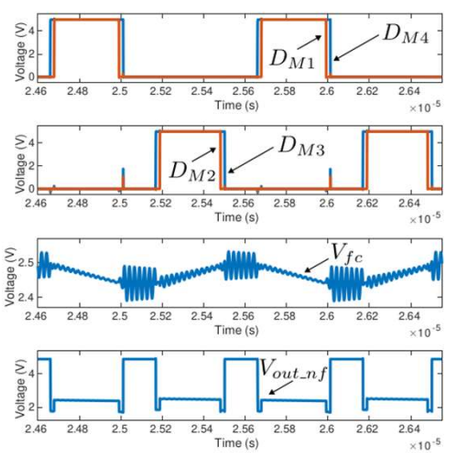
This workshop's plan

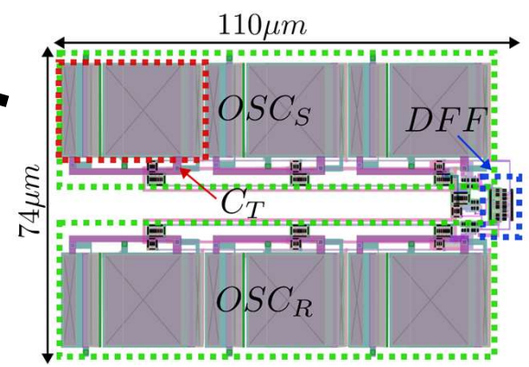




DAY 1

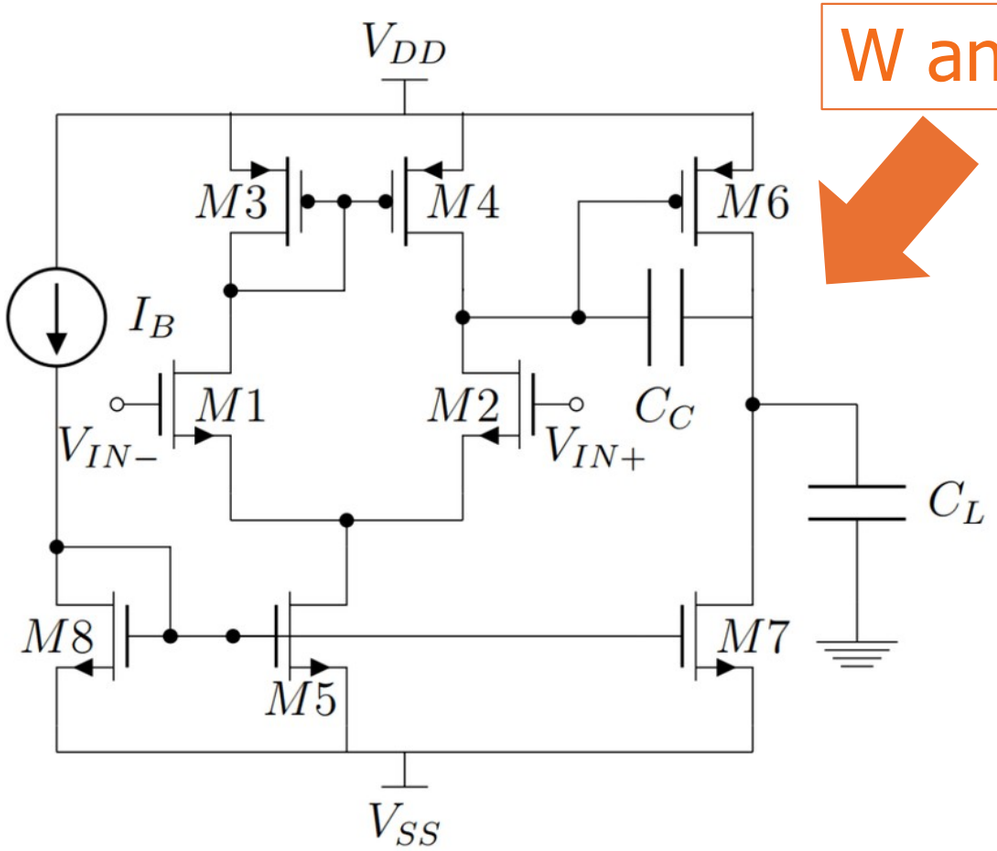
DAY 2



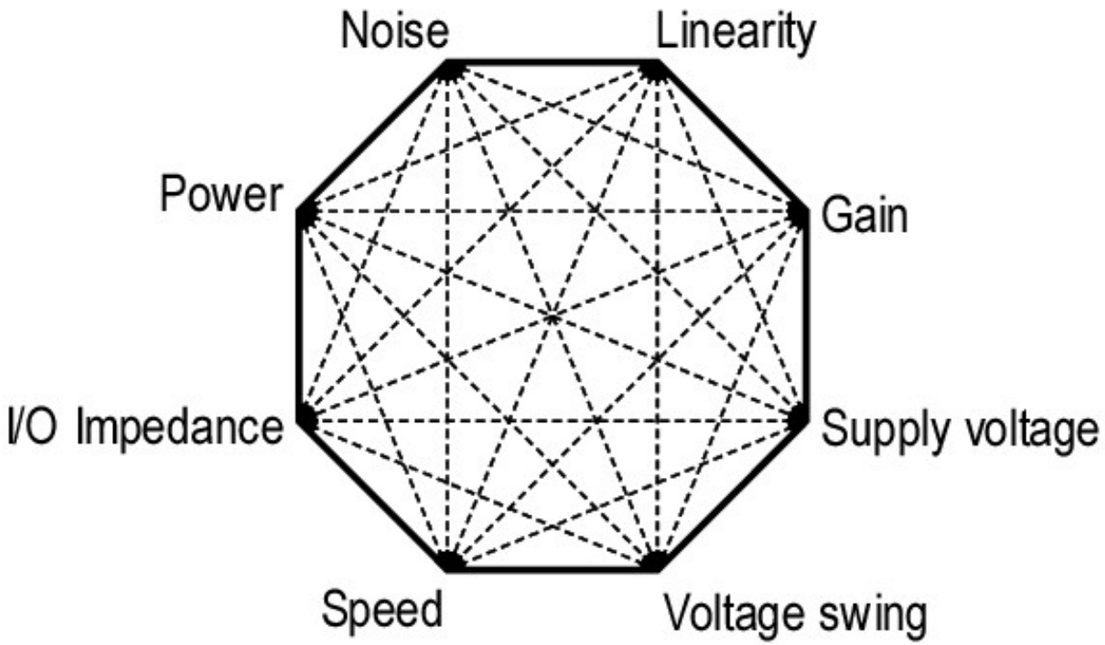


Open-Source Analog Design

Classical tradeoffs in analog design



W and L for each transistor M1 to M8?

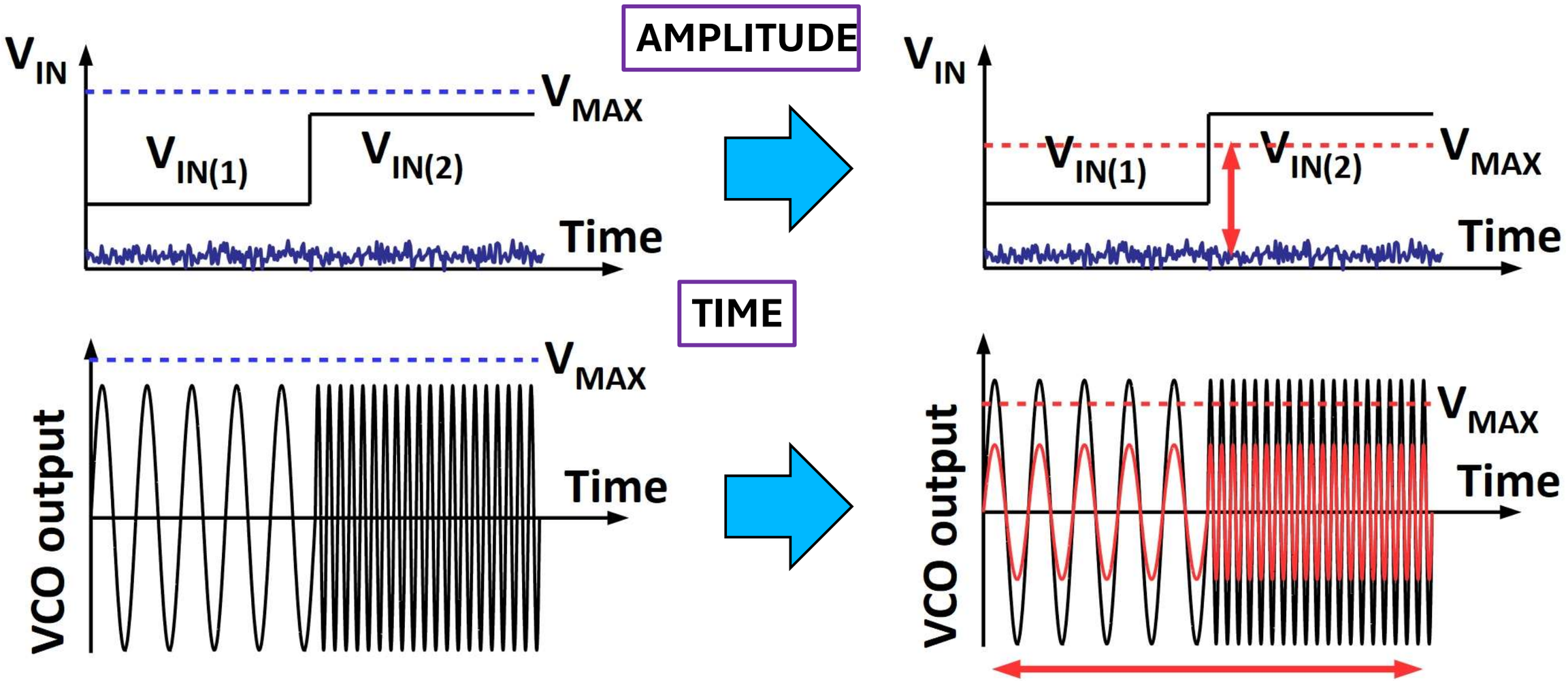


Analog block example: Miller OTA

[B. Razavi]

Performance is limited by the tradeoff in target specifications

From analog- to time-based processing

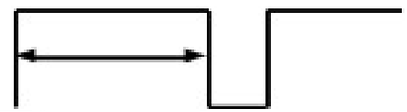


Amplitude- vs time-based circuits



Amplitude-based

- Well-known architectures and techniques
- SNR reduced with voltage headroom
- Area consuming
- Sensible to perturbations (analog blocks)

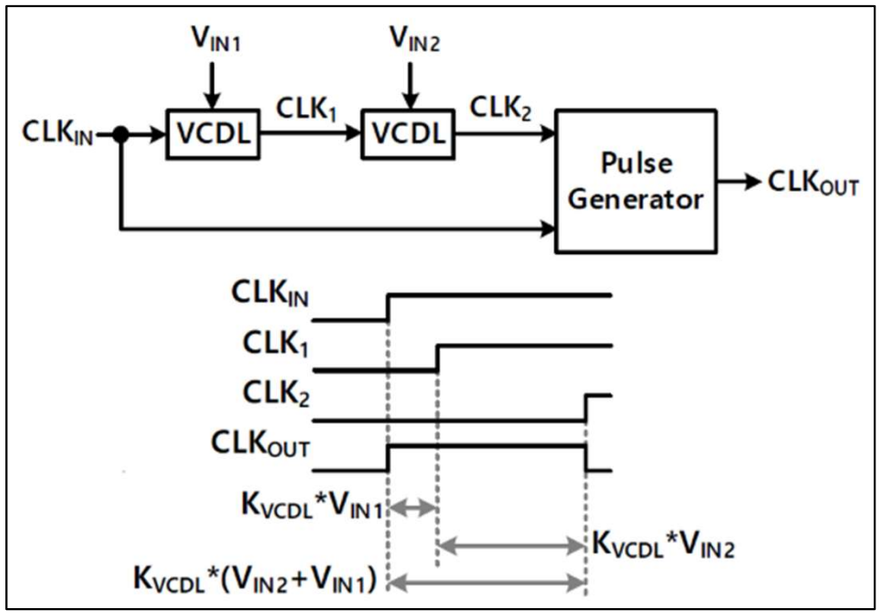


Time-based

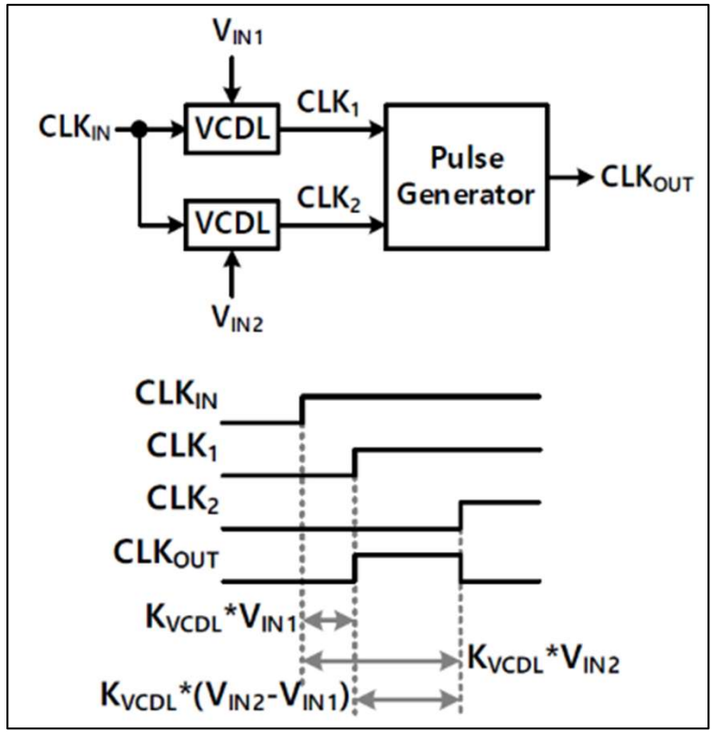
- Unlimited headroom in time domain
- Compact area
- Robust (highly-digital)
- New concepts need to be developed
- Highly-nonlinear V-to-f conversion

Theoretical power efficiency is comparable in both cases [Van Rethy AACD2014]

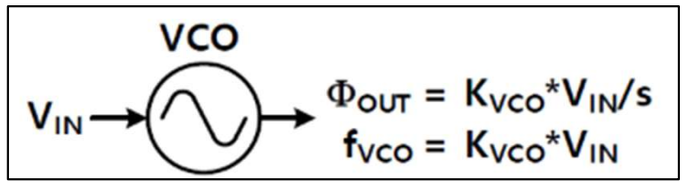
Time-based function implementation



(a) Addition

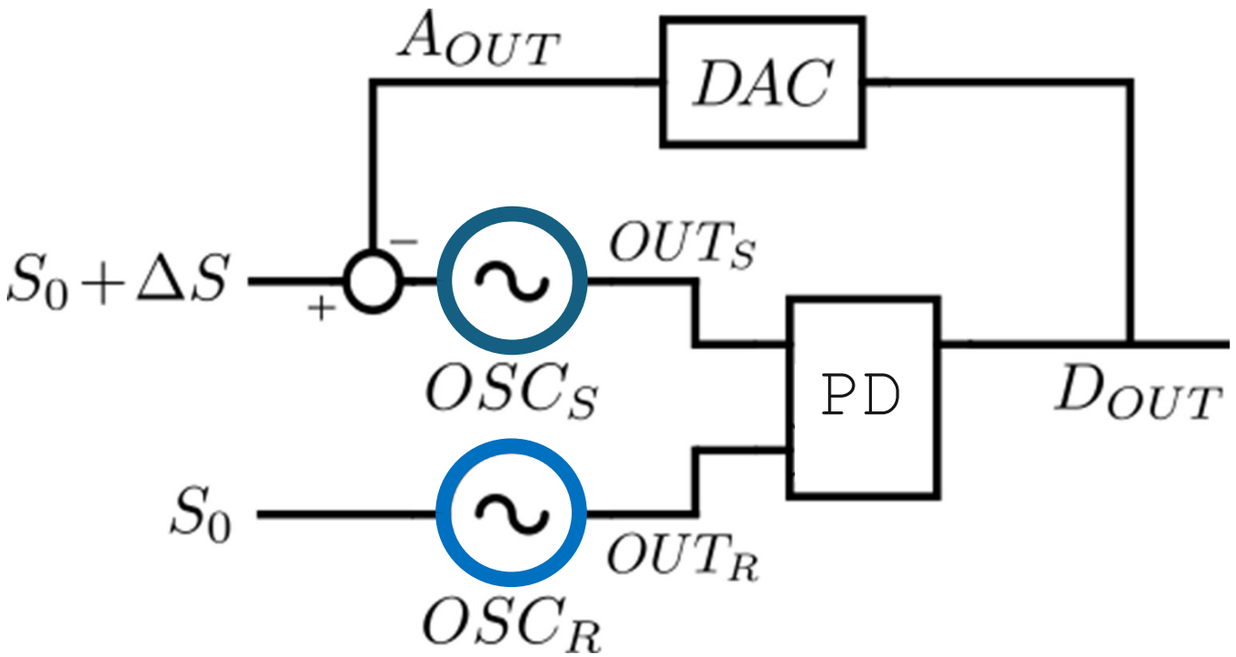


(b) Subtraction



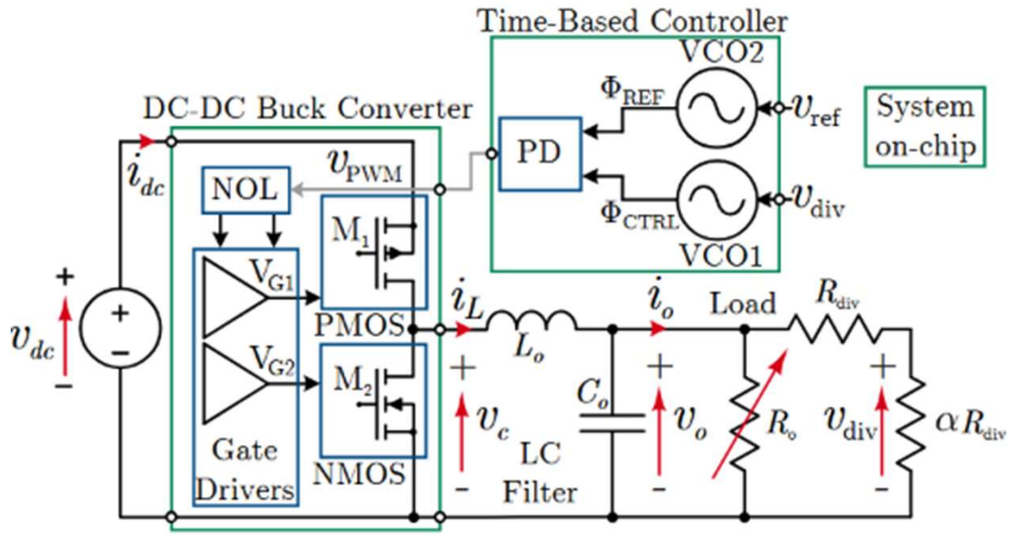
(b) Integration

Recent examples



$$\overline{D_{OUT}} = 1 - C_S/\Delta C$$

[Marin, LASCAS 2023]



[Marin, NORCAS 2025]

Schematic design and simulation

- Relevant tools
  - Xschem → schematic entry and netlist generation
  - Ngspice → simulation based on netlist generated by Xschem

- Visualization
  - Ngspice window → quick checks
  - GAW → integrated in Xschem
  - External viewer through raw data (e.g. Python script)

Simulation types

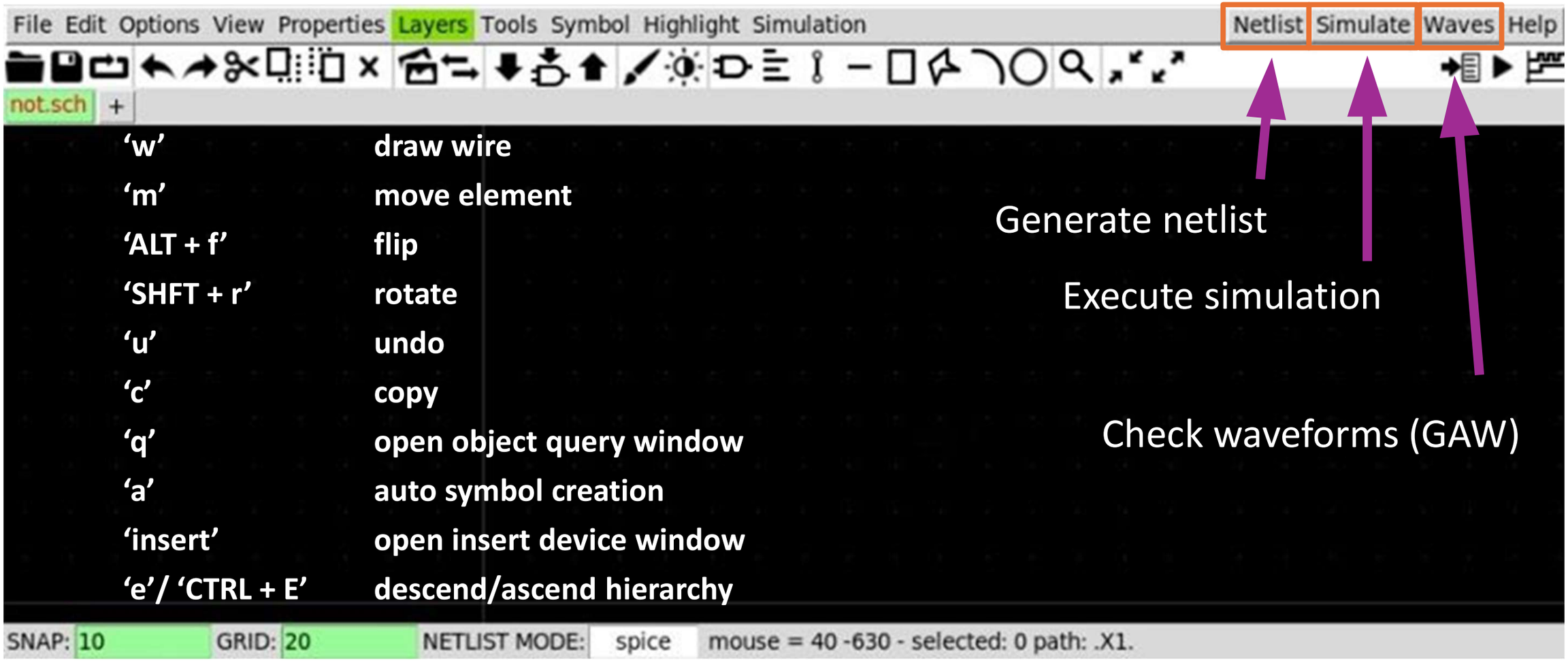
- DC → analysis of the circuit operating point
- Transient → time-domain behaviour
- AC → frequency sweep
- Noise → simulation of device intrinsic noise
- And others…

Ngspice manual will become your best friend!
https://ngspice.sourceforge.io/docs/ngspice-manual.pdf



Xschem GUI basics part 1

www.xschem.sourceforge.io



Simulation scope

- Nominal
  Ideal simulation without considering many fabrication effects
- P(VT) corners
  Considers global process variation (P) and environment (Voltage, temperature, T)

```
.lib cornerMOSlv.lib mos_tt
.lib cornerRES.lib res_typ
.lib cornerCAP.lib cap_typ
```

- Mismatch
  Considers local statistical variation among devices
  → See examples in /opt/pdks/ihp-sg13g2/libs.tech/xschem/sg13g2_tests/mc_*.sch
- Parasitic extraction/ post layout simulations
  Components associated to extrinsic structures (metallization)

Simulation examples

Set repo working folder

- OUTSIDE Docker:
  → open [YOUR_INSTALL_FOLDER]\ac3e-ic-toolbox\.env
    → Set path as you want
      → In my case: [DESIGN_FOLDER] = C:\Users\EQUIPO\Documents\EIE-PUCV\Associate_professor\Simulation\designs
    → Example: SHARED_DIR=/mnt/c/Users/EQUIPO/Documents/EIE-PUCV/Associate_professor/Simulation/designs
- INSIDE Docker
  - Go to /home/designer/shared
  - Check if files are seen here (try with workshop repo, see next slide)

Clone Workshop repo

- Set PDK to IHP 130nm → INSIDE Docker:
  → ls /opt/pdks/ → see installed PDKs
  → set_pdk ihp-sg13g2

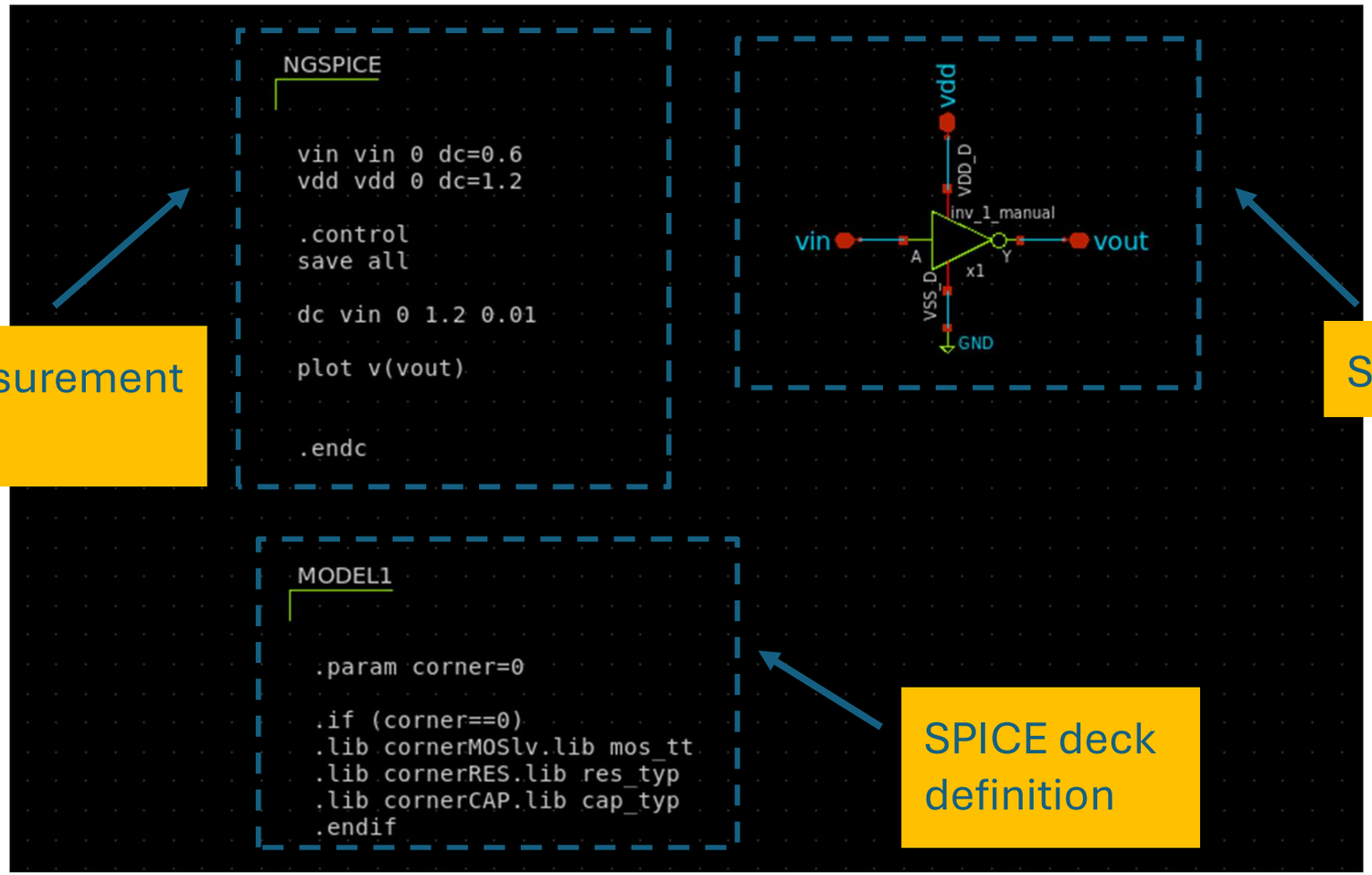
```
designer ~
$ set_pdk ihp-sg13g2
PDK set to ihp-sg13g2
```

- OUTSIDE Docker:
  → cd [DESIGN_FOLDER]
  → git clone https://github.com/JorgeMarinN/OS_AnalogIC_UFRO-UBB_Nov2026

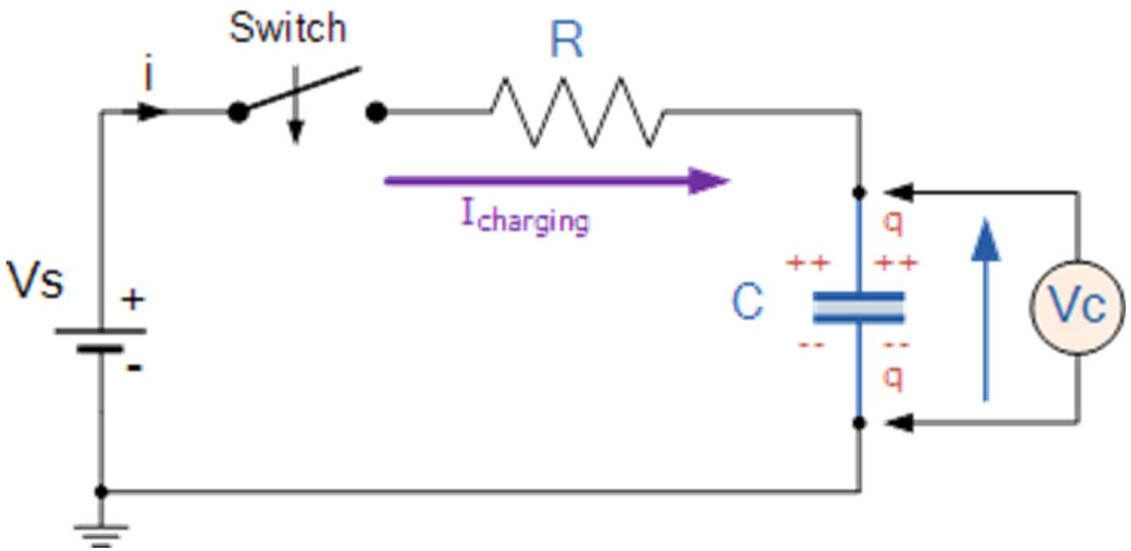
Example #1: Basic inverter

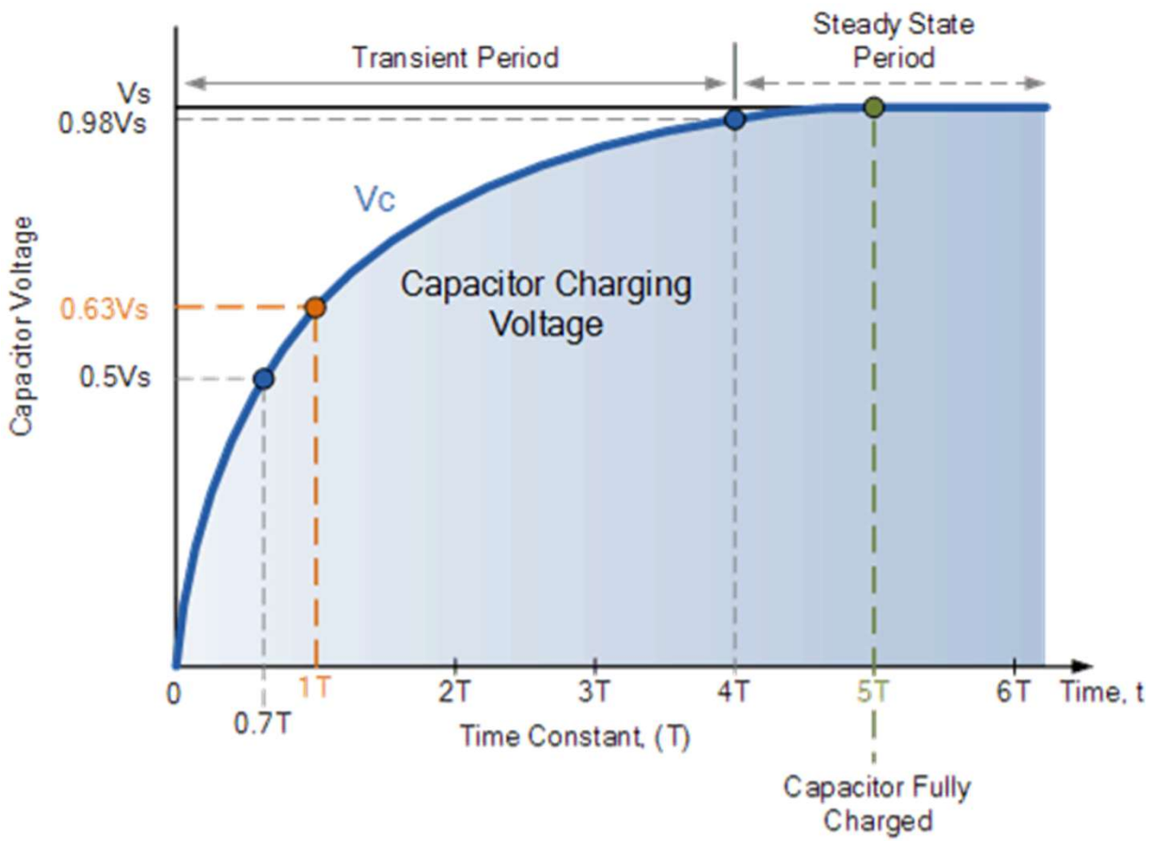
- Github link:

→ https://github.com/JorgeMarinN/OS_AnalogIC_UFRO-UBB_Nov2026/blob/main/Day1/tb_inv_1_manual.sch



Example #2: RC constant calculation





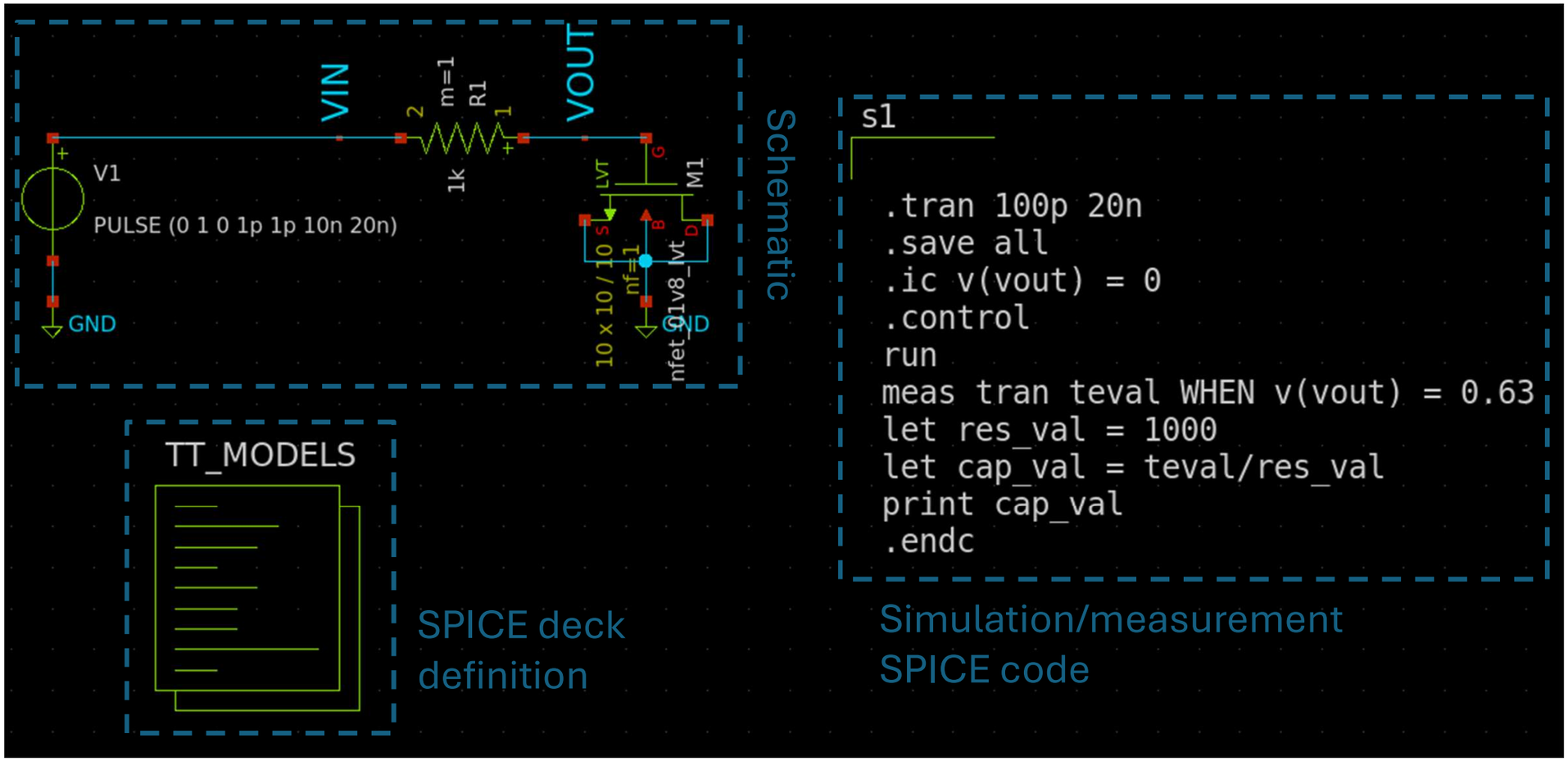
Source: https://www.electronics-tutorials.ws/rc/rc_1.html

Example #2: RC constant calculation

- Github link:

→ https://github.com/JorgeMarinN/OS_AnalogIC_UFRO-UBB_Nov2026/blob/main/Day1/tb_rc-ext_circuit.sch

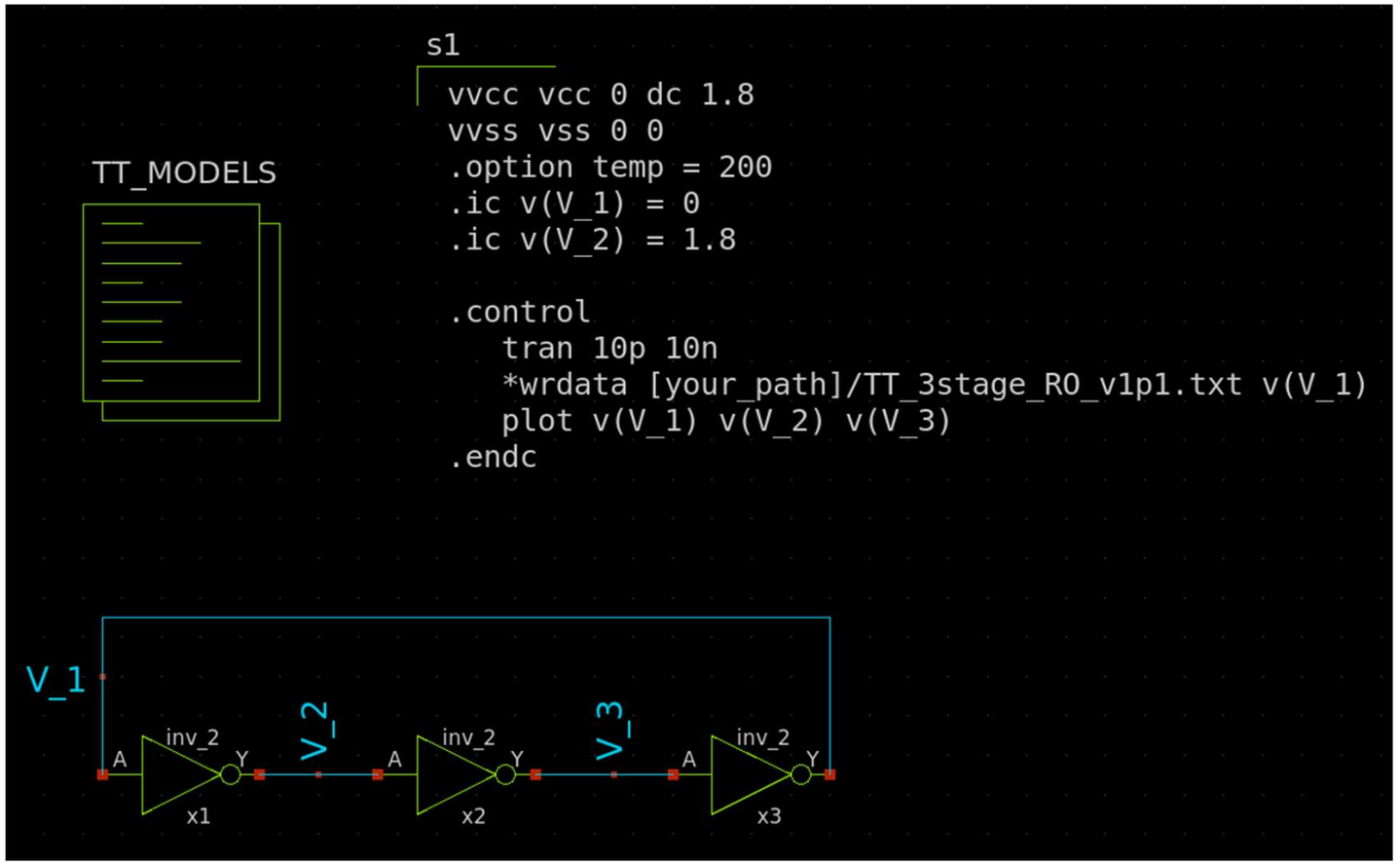
→ https://github.com/JorgeMarinN/OS_AnalogIC_UFRO-UBB_Nov2026/blob/main/Day1/tb_cgate-ext_circuit.sch



Example #3: ring oscillator

- Github link:

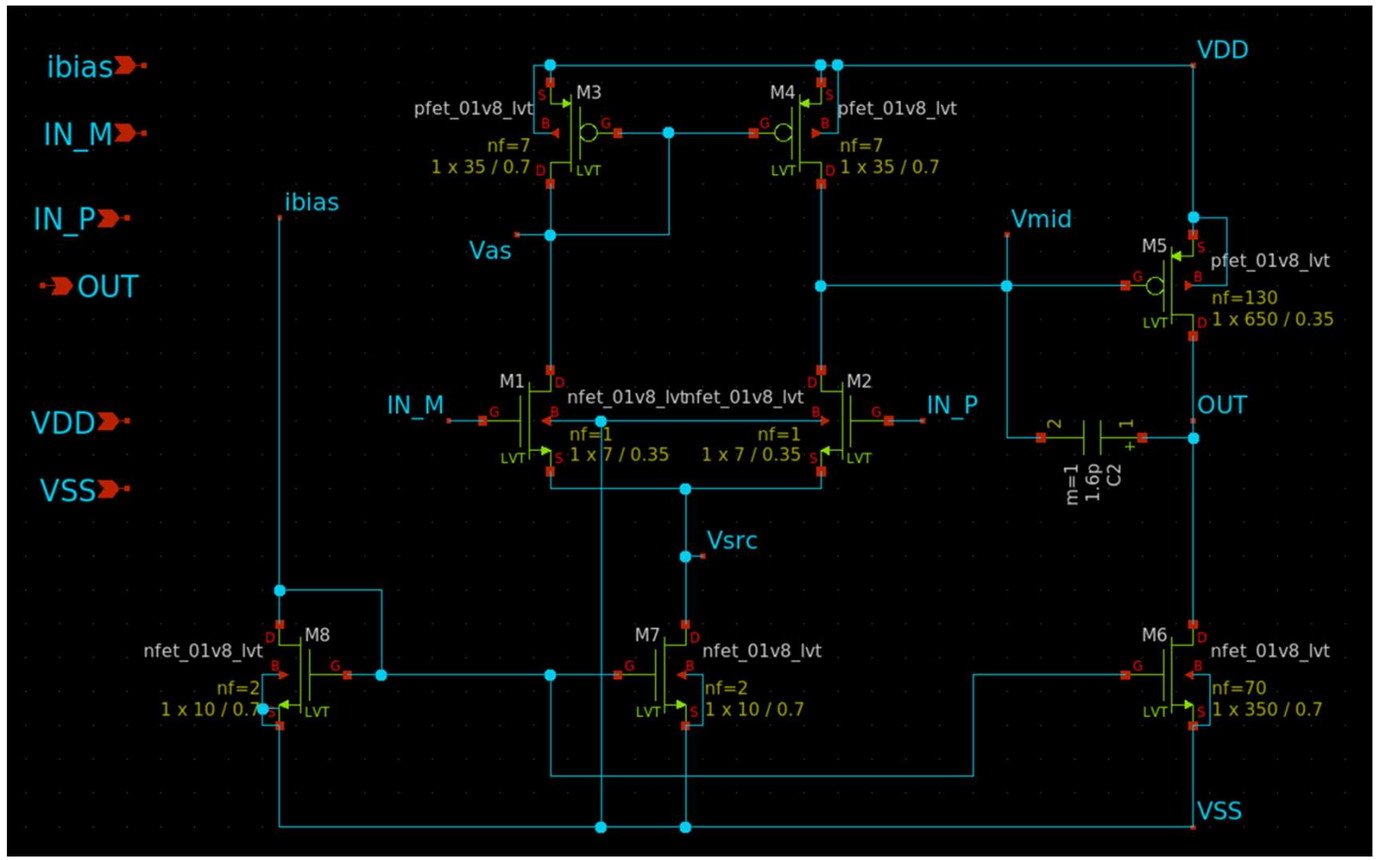
→https://github.com/JorgeMarinN/OS_AnalogIC_UFRO-UBB_Nov2026/blob/main/Day1/tb_3stage_RO.sch



Proposed activities

Activity	Proposed task and expected result
Design a Voltage-Controlled Oscillator (VCO)	Introduce a voltage-controlled delay mechanism (e.g., variable supply voltage or a controlled delay stage). Sweep the control voltage and plot f_{osc} versus V_{ctrl}. Estimate VCO gain K_{VCO} in MHz/V.
Design a 100 MHz Ring Oscillator	Modify the number of inverter stages to obtain approximately 100 MHz. Estimate the silicon area from the standard-cell dimensions and compare frequency, area, and power against the original oscillator.
Explore PVT Variations	Simulate the oscillator at different process corners (TT, SS, FF), temperatures (−40, 27, 100 °C), and supply voltages (1.08, 1.2, 1.32 V). Plot the frequency variation and identify the worst-case conditions.

Example #4: Miller OTA simulation



$$GBW = \frac{gm_{1,2}}{2\pi C_c}$$

$$fnd = \frac{gm_5}{2\pi C_L} \cdot \frac{1}{1 + \frac{C_{n1}}{C_c}} \quad > \mathbf{3GBW}$$

Two equations for

Three variables g_{m1}, g_{m6} and C_c ?!?

Solution : choose g_{m1} or g_{m6} or C_c !!!

[W. Sansen, Analog Design Essentials]

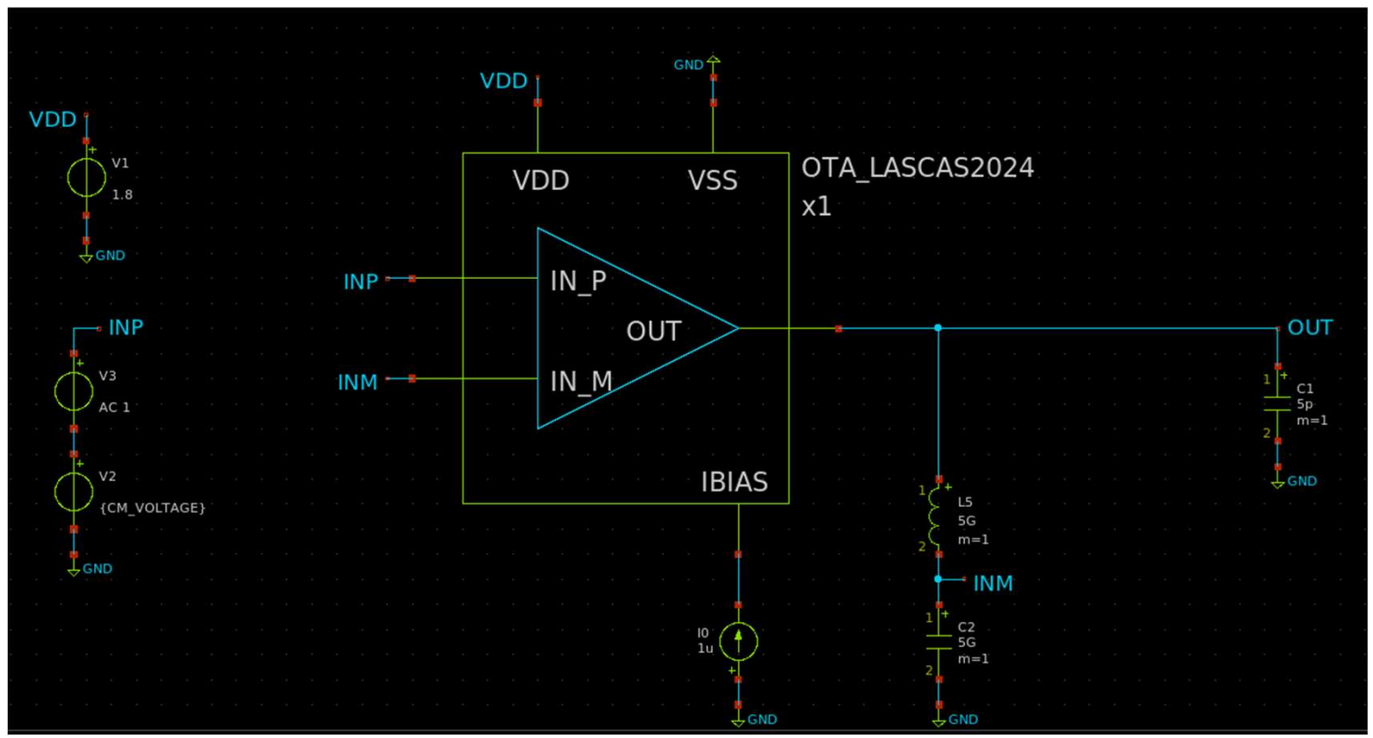
Example #4: Miller OTA simulation

- Github link:

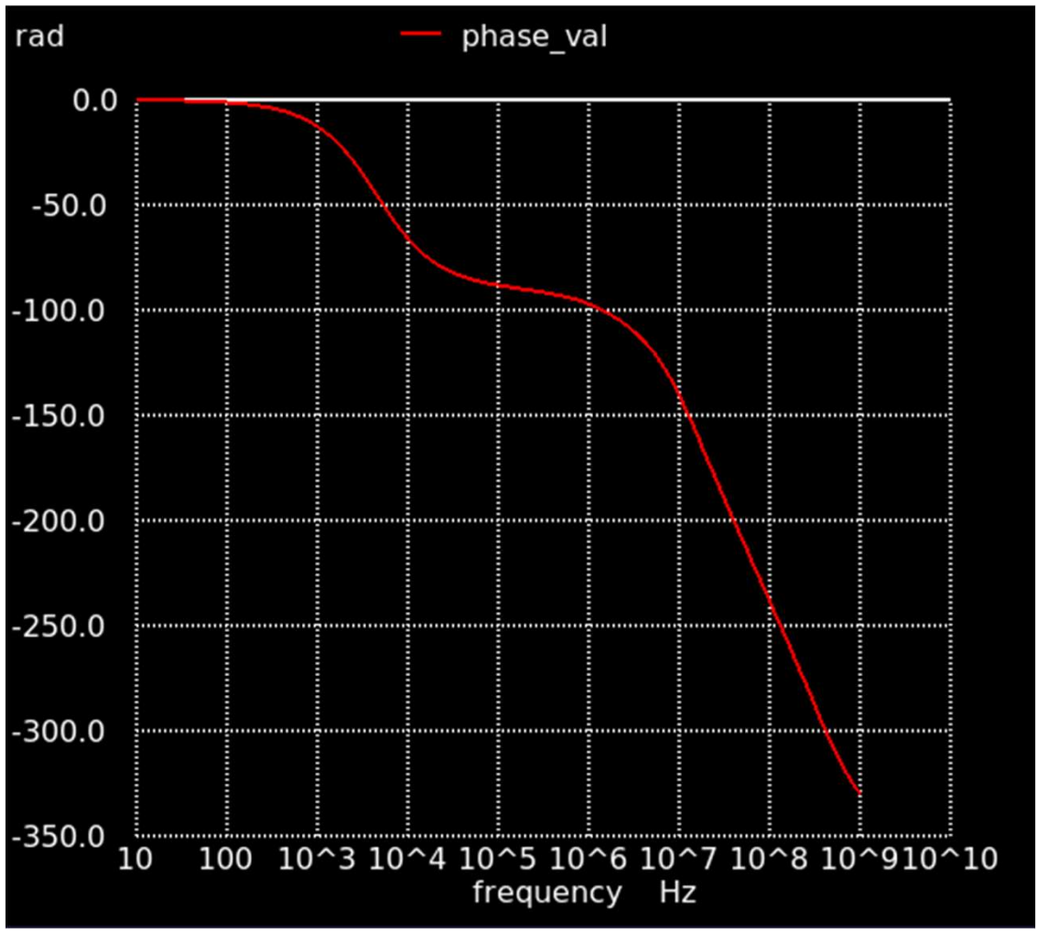
→https://github.com/JorgeMarinN/UNIC-CASS2024_AnalogOSIC/blob/main/tb_OTA_stability_UNIC-CASS2024.sch

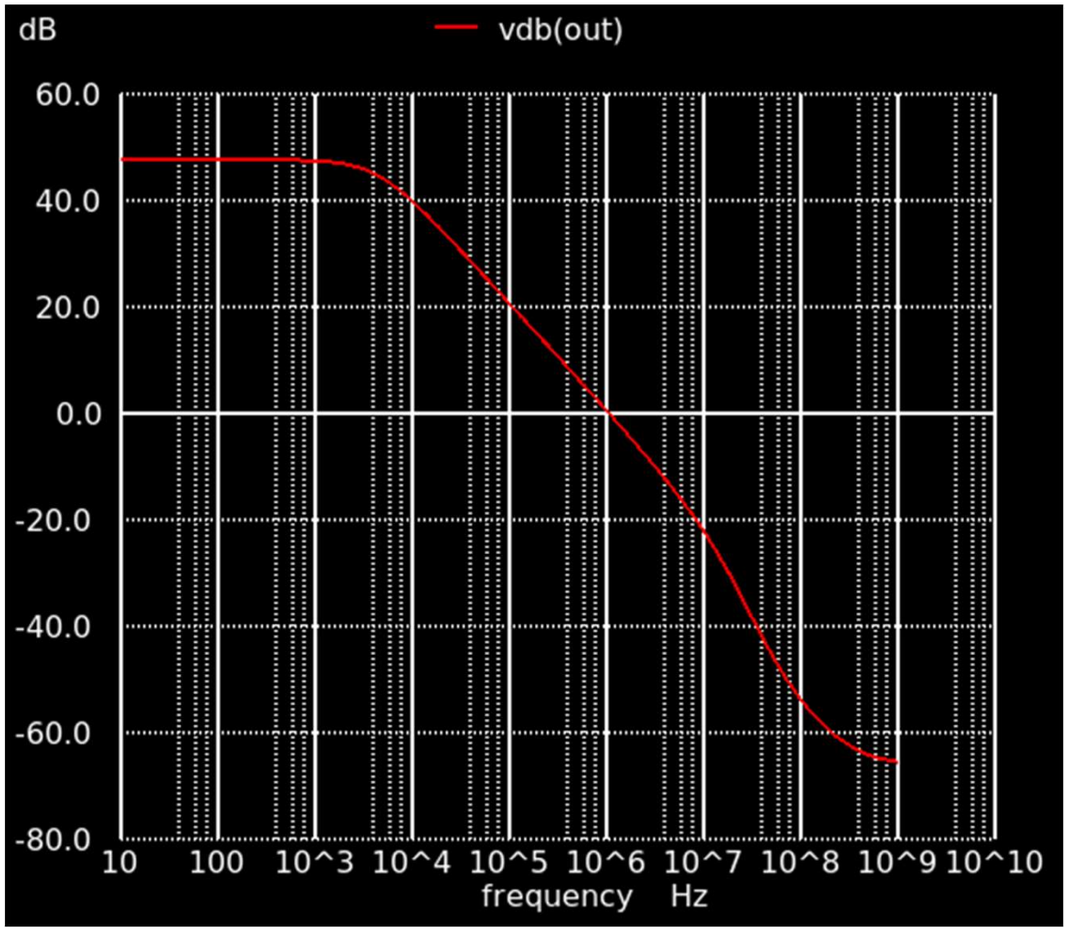
NGSPICE

```
.param CM_VOLTAGE = 0.9
.control
save all
ac dec 200 10 1000Meg
settype decibel out
plot vdb(out)
let phase_val = 180/PI*cph(out)
let PM_val = 180 + 180/PI*cph(out)
settype phase phase_val
plot phase_val
meas ac PM FIND PM_val WHEN vdb(out)=0
meas ac GBW WHEN vdb(out)=0
op
```



Example #4: Miller OTA simulation





pm	=	8.248775e+01
gbw	=	1.058205e+06

Education using AMS design OS tools

- Traditional IC design flows are not designed for educational purposes
  - The wide variety of tools and workflows can be overwhelming for beginners
  - High costs and restricted PDK availability limit accessibility
  - Limited documentation and minimal community-driven collaboration hinder learning
- Tapeout courses require significant budgets, making them inaccessible to many universities and regions
- IC design environments have traditionally been restrictive
  - Strict NDAs restrict access to essential resources
  - Information sharing is trust-based and highly limited
  - However, this landscape is rapidly evolving, opening new opportunities for education!

Removing barriers: knowledge base

- JKU's "Analog Circuit Design course" → https://github.com/iic-jku/analog-circuit-design
  - Fully-open content for intermediate-level MOSFET circuit design course
  - Based on Xschem and Ngspice examples + IHP technology
  - Uses JKU's IIC-OSIC-TOOLS Docker
- University of Hawai'i's "Analysis and Design of Integrated Circuits" tapeout course → https://github.com/bmurmann/EE628
  - Open content for lectures and assignments
  - Design of a Sigma-delta converter using OS design flow + IHP technology

Removing barriers: knowledge base

- MANY other sources:
  - Carsten Wulff's course: https://www.youtube.com/playlist?list=PLybHXZ9FyEhYuu3SJjXBncbR3BEImeZ0H
  - Angel Abusleme's course [in Spanish]: https://youtube.com/playlist?list=PLDYu8HgBbvRFf2PahRmg4ABDxRN60ujKQ&si=jnnEGgbIed8ik8av
  - Tiny tapeout: https://tinytapeout.com/specs/analog/
  - IHP Analog Academy: https://github.com/IHP-GmbH/IHP-AnalogAcademy
  - Harald Pretl's course: https://github.com/iic-jku/hpretl-lectures
  - Kwantae Kim's course: https://www.youtube.com/playlist?list=PLmK06TkPUUKo-2LmZfhBpbcLTjFOxk2WJ

Thank you!
Questions?